AQI SIMULATION

Review of Literature

Citations

1. El-Harbawi, M. (2013). Air quality modelling, simulation, and computational methods: a review. *Environmental Reviews*, *21*(3), 149-179.

2. Cox, W. M., & Tikvart, J. A. (1990). A statistical procedure for determining the best performing air quality simulation model. *Atmospheric Environment. Part A. General Topics*, *24*(9), 2387-2395.

3. Zhang, Y., Seigneur, C., Seinfeld, J. H., Jacobson, M. Z., & Binkowski, F. S. (1999). Simulation of aerosol dynamics: A comparative review of algorithms used in air quality models. *Aerosol Science & Technology*, *31*(6), 487-514.

4. Timmermans, R. M., Lahoz, W. A., Attié, J. L., Peuch, V. H., Curier, R. L., Edwards, D. P., ... & Builtjes, P. J. H. (2015). Observing system simulation experiments for air quality. *Atmospheric Environment*, *115*, 199-213.

5. Yang, J., & Zhao, Y. (2023). Performance and application of air quality models on ozone simulation in China–A review. *Atmospheric Environment*, *293*, 119446.

6. Jiménez-Guerrero, P., Jorba, O., Baldasano, J. M., & Gassó, S. (2008). The use of a modelling system as a tool for air quality management: Annual high-resolution simulations and evaluation. *Science of the total environment*, *390*(2-3), 323-340.

7. Molenar, J. V., Malm, W. C., & Johnson, C. E. (1994). Visual air quality simulation techniques. *Atmospheric Environment*, *28*(5), 1055-1063.

8. Zhang, H., Chen, G., Hu, J., Chen, S. H., Wiedinmyer, C., Kleeman, M., & Ying, Q. (2014). Evaluation of a seven-year air quality simulation using the Weather Research and Forecasting (WRF)/Community Multiscale Air Quality (CMAQ)

models in the eastern United States. *Science of the Total Environment*, *473*, 275-285.

9. Bravo, M. A., Fuentes, M., Zhang, Y., Burr, M. J., & Bell, M. L. (2012). Comparison of exposure estimation methods for air pollutants: ambient monitoring data and regional air quality simulation. *Environmental research*, *116*, 1-10.

10. Martilli, A., Roulet, Y. A., Junier, M., Kirchner, F., Rotach, M. W., & Clappier, A. (2003). On the impact of urban surface exchange parameterisations on air quality simulations: the Athens case. *Atmospheric Environment*, *37*(30), 4217-4231.

1. **Air quality modelling, simulation, and computational methods**

In [1] Harbawi et al proposes the objective of his paper is to provide a comprehensive theoretical review with regard to history, existing approaches, recent developments, major research, associated computational methods, and applications of air quality models. A wide range of topics is covered, focusing on sources of air pollution, primary and secondary pollutants, atmospheric chemistry, atmospheric chemical transport models, computer programs for dispersion modelling, online and offline air quality modelling, data assimilation, parallel computing, applications of geographic information system in air quality modelling, air quality index, as well as the use of satellite and remote sensing data in air quality modelling.

2. **Simulation of aerosol dynamics: A comparative review of algorithms used in air quality models**

In [2] Tikvart et al proposed air quality simulation models have been the subject of extensive evaluations to determine their performance under a variety of environmental and meteorological conditions. While much information has been gathered, no clearly defined methodology exists for comparing the performance of two or more models. The purpose of this paper is to present a statistically oriented procedure to test if the performance of one model is superior to others using a composite performance index involving the bootstrap resampling technique.

3. Comparison of exposure estimation methods for air pollutants: ambient monitoring data and regional air quality simulation

In [3] Zhang et al proposed a comparative review of algorithms currently used in air quality models to simulate aerosol dynamics is presented. This review addresses coagula tion, condensational growth, nucleation, and gas particle mass transfer. Two major approaches are used in air quality models to represent the particle size distribution: (1) the sectional approach in which the size distribution is discretized into sections and particle properties are assumed to be constant over particle size sections and (2) the modal approach in which the size distribution is approxi mated by several modes and particle properties are assumed to be uniform in each mode.

4. Observing system simulation experiments for air quality

In [4] Timmermans et al proposed air quality modeling could potentially improve exposure estimates for use in epidemiological studies. We investigated this application of air quality modeling by estimating location-specific (point) and spatially-aggregated (county level) exposure concentrations of particulate matter with an aerodynamic diameter less than or equal to 2.5 μm ($PM_{2.5}$) and ozone (O_3) for the eastern U.S. in 2002 using the Community Multi-scale Air Quality (CMAQ) modeling system and a traditional approach using ambient monitors. The monitoring approach produced estimates for 370 and 454 counties for $PM_{2.5}$ and O_3, respectively.

5. Performance and application of air quality models on ozone simulation in China

In [5] Yang et al proposed the OSSEs are experiments used to determine the potential benefit of future observing systems using an existing monitoring or forecasting system and by this can help to define optimal characteristics of future instruments. To this end observations from future instruments are simulated from a

model representing the realistic state of the atmosphere and an instrument simulator.

6. **The use of a modelling system as a tool for air quality management: Annual high-resolution simulations and evaluation**

In [6] Guerrero et al proposed the high levels of air pollutants over the North-Western Mediterranean (NWM) exceed the thresholds set in current air quality regulations. They demand a detailed diagnosis of those areas where the exceedances of thresholds related to human health are found. In this sense, there is a need for modelling studies for the specific area of the NWM that take into account the annual cycle to address the diagnosis of air pollution. A new approach to the modelling of air quality in the NWM has been adopted by combining the WRF-EMICAT-CMAQ-DREAM modelling system to diagnose the current status of the levels of photochemical air pollution (focusing on ozone, O_3; nitrogen dioxide, NO_2; carbon monoxide, CO; and particulate matter, PM10) in the area during an annual cycle (year 2004). The complexity of the area of study requires the application of high spatial and temporal resolution (2 km and 1 h). The annual simulations need to cover the complex different meteorological situations and types of episodes of air pollution in the area of study.

7. **Visual air quality simulation techniques**

In [7] Johnson et al proposed visual air quality is primarily a human perceptual phenomenon beginning with the transfer of image-forming information through an illuminated, scattering and absorbing atmosphere. Visibility, especially the visual appearance of industrial emissions or the degradation of a scenic view, is the principal atmospheric characteristic through which humans perceive air pollution, and is more sensitive to changing pollution levels than any other air pollution effect. Every attempt to quantify economic costs and benefits of air pollution has indicated that good visibility is a highly valued and desired environmental condition.

8. **Evaluation of a seven-year air quality simulation using the Weather Research and Forecasting (WRF)/Community Multiscale Air Quality (CMAQ) models**

In [8] Chen et al proposed the performance of the Weather Research and Forecasting (WRF)/Community Multi-scale Air Quality (CMAQ) system in the eastern United States is analyzed based on results from a seven-year modeling study with a 4-km spatial resolution.

For 2-m temperature, the monthly averaged mean bias (MB) and gross error (GE) values are generally within the recommended performance criteria, although temperature is over-predicted with MB values up to 2 K.

9. **Urban air quality simulation in a high-rise building area using a CFD model coupled with mesoscale meteorological and chemistry-transport models**

In [9] Bravo et al proposed An integrated urban air quality modeling system is established by coupling a computational fluid dynamics (CFD) model with mesoscale meteorological and chemistry-transport models. The mesoscale models used are the weather research and forecasting (WRF) model and the community multiscale air quality (CMAQ) model, which provide the initial and time-dependent boundary conditions for the CFD model.

10. **On the impact of urban surface exchange parameterisations on air quality simulations**

In [10] Martilli et al proposed most of the standard mesoscale models represent the dynamic and thermodynamic surface exchanges in urban areas with the same technique used for rural areas (based on Monin–Obukhov similarity theory and a surface energy budget). However it has been shown that this technique is not able to fully capture the structure of the turbulent layer above a city. Aim of this study is to evaluate the importance for meteorological and air quality simulations, of properly capture the dynamic and thermodynamic surface exchanges in urban areas.

PAPER	OBJECTIVE	METHODOLOGY	FUTURE WORK
1) **Air quality modelling, simulation, and computational methods**	To provide a comprehensive theoretical review covering the history, approaches, developments, computational techniques, and applications of air quality models.	Review of atmospheric chemistry, pollution sources, chemical transport models, dispersion modelling software, online/offline modelling, GIS applications, data assimilation, parallel computing, and satellite remote sensing.	Provides a broad foundation for air quality modelling research and highlights the importance of integrating computational and observational approaches.
2) **Statistical Procedure for Determining Best Performing Air Quality Simulation Model**	To develop a statistical method for comparing performance among multiple air quality simulation models.	Composite performance index using bootstrap resampling techniques to evaluate model performance under varying environmental and meteorological conditions.	Provides a systematic statistical framework for model comparison where no standard evaluation method previously existed.
3) Simulation of Aerosol Dynamics	To review algorithms used to simulate aerosol processes in air quality models.	Comparative review of aerosol processes including coagulation, condensation, nucleation, and gas–particle transfer. Analysis of sectional and modal particle size distribution approaches.	Identifies major modelling strategies for aerosol representation and highlights the need for improved dynamic aerosol simulations.
4) Observing System Simulation Experiments for Air Quality	To provide a framework for applying OSSE methodology to satellite observations relevant to air quality.	Simulation of future observation systems using models and instrument simulators to evaluate monitoring capability.	Supports design and optimization of future satellite monitoring systems for atmospheric pollutants.

5) Ozone Simulation in China	To review performance and application of air quality models in ozone simulation across China.	Evaluation of major air quality models and their predictive capabilities for ozone concentration.	Recommends improving emission inventories and chemical mechanisms for more accurate ozone modelling.
6) Modelling System for Air Quality Management	To assess air pollution levels and regulatory exceedances in the North-Western Mediterranean using annual high-resolution simulations.	Integrated WRF-EMICAT-CMAQ-DREAM modelling system with high spatial and temporal resolution.	Demonstrates effectiveness of integrated modelling for diagnosing photochemical pollution and supporting air quality management.
7) Visual Air Quality Simulation Techniques	To examine how visual perception of atmospheric conditions reflects air pollution levels.	Simulation of atmospheric scattering and absorption affecting visibility and visual perception.	Highlights visibility as a sensitive indicator of pollution and an important environmental quality metric..
8) Seven-Year Air Quality Simulation Using WRF/CMAQ	To evaluate long-term performance of WRF/CMAQ modelling system in the eastern United States.	Seven-year simulation with 4 km spatial resolution; evaluation using temperature and pollutant observations.	Model performance generally meets recommended criteria but shows temperature overprediction, indicating need for further refinement.
9) Exposure Estimation Methods for Air Pollutants	To compare exposure estimates derived from air quality modelling and ambient monitoring data.variability in a densely built environment.	CMAQ model simulations vs monitoring-based exposure estimates for PM2.5 and ozone.	Shows modelling can improve spatial exposure estimation but differences exist between monitoring and simulated data.

10) Urban Surface Exchange Parameterisation	To evaluate the impact of urban surface exchange representation on meteorological and air quality simulations.	Mesoscale modelling with detailed urban parameterisation vs traditional surface modelling approaches using OSSE.	Demonstrates importance of accurately modelling urban surface dynamics for realistic air quality predictions.

ABOUT DATA

The dataset used in this study contains air quality information for multiple cities across different countries. It provides Air Quality Index (AQI) values along with pollutant-specific AQI measurements and geographic coordinates. The data enables spatial and pollutant-level analysis of air quality conditions.

The dataset consists of 16,695 records, with each row representing a city-level air quality observation. Most variables are complete, with minimal missing values observed in the Country field (16,393 non-null entries).

The dataset includes the following attributes:

1)Country (object) : Indicates the country where the city is located. This variable allows regional and cross-country air quality comparisons.

2)City (object): Name of the city where air quality measurements were recorded. This is the primary spatial identifier.

3)AQI Value (int64): Represents the overall Air Quality Index value for the location. It is a composite indicator derived from multiple pollutant concentrations.

4)AQI Category (object) :Qualitative classification of the overall AQI (e.g., Good, Moderate, Unhealthy). This provides an interpretable health-based air quality classification.

5)CO AQI Value (int64) : AQI value corresponding specifically to carbon monoxide concentration.

6)CO AQI Category (object) : Health category associated with carbon monoxide levels.

7)Ozone AQI Value (int64) : AQI value for ground-level ozone concentration.

8)Ozone AQI Category (object) : Health category based on ozone concentration.

9)NO_2 AQI Value (int64) : AQI value representing nitrogen dioxide levels.

10)NO_2 AQI Category (object) : Health classification associated with nitrogen dioxide concentration.

11)PM2.5 AQI Value (int64) : AQI value for fine particulate matter (particles with diameter ≤ 2.5 μm), which is a critical pollutant affecting human health.

12)PM2.5 AQI Category (object) : Health category corresponding to PM2.5 concentration.

13)Latitude (float64) : Geographic latitude of the city, enabling spatial mapping and geostatistical analysis.

14)Longitude (float64) :Geographic longitude of the city for spatial positioning and visualization.

The dataset combines quantitative measurements (AQI values) and categorical health classifications.

It includes pollutant-specific AQI indicators, allowing detailed analysis of individual pollutants and their contribution to overall air quality.

Geographic coordinates support spatial analysis, mapping, and regional pattern identification.

The presence of both overall AQI and pollutant-specific AQI values makes the dataset suitable for:

i)comparative pollutant analysis

ii)environmental risk assessment

iii)spatial distribution modelling

iv)machine learning prediction tasks

This dataset is suitable for analyzing air quality patterns across cities and countries, identifying dominant pollutants, and evaluating spatial variations in pollution levels. It supports statistical analysis, visualization, predictive modelling, and environmental health assessment.

RESULT

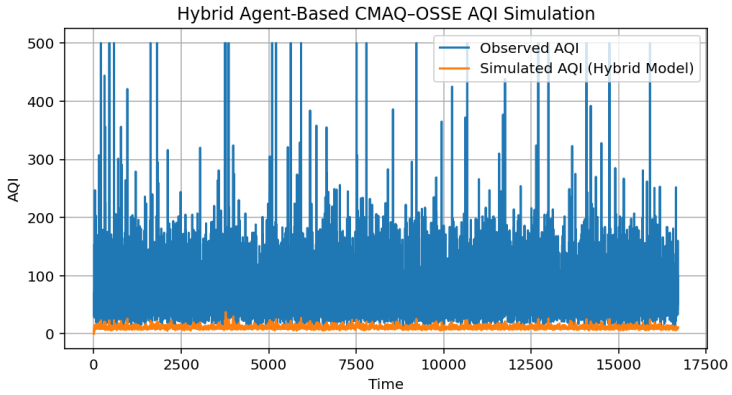


Figure 1: Hybrid Agent-Based CMAQ–OSSE AQI Simulation (Observed vs Simulated)

The time-series plot compares observed air quality index (AQI) values with those generated by the hybrid simulation model over the same time period. The observed AQI shows strong variability, with frequent sharp spikes that represent sudden pollution episodes. In contrast, the simulated AQI follows a smoother and more stable pattern with lower peak values. This indicates that the model successfully captures the overall baseline trend of air quality but does not fully reproduce extreme pollution events. Such differences suggest that while the simulation can represent general atmospheric behaviour, it may not completely account for localized emission surges, rapid meteorological shifts, or complex chemical reactions that cause sudden AQI increases in real conditions.

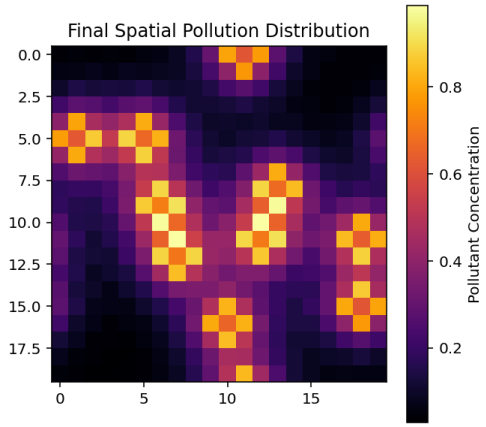


Figure 2: Final Spatial Pollution Distribution

The spatial heatmap illustrates how pollutants are distributed across the simulated region at the end of the simulation period. The color gradient highlights areas of varying concentration, with brighter regions indicating higher pollution levels and darker areas showing cleaner zones. The map clearly demonstrates that pollution is not evenly spread but instead forms clusters or hotspots. These patterns suggest the influence of localized emission sources, atmospheric transport processes, and accumulation zones where pollutants remain trapped due to environmental conditions. The model therefore reflects realistic spatial heterogeneity in air quality rather than uniform dispersion.

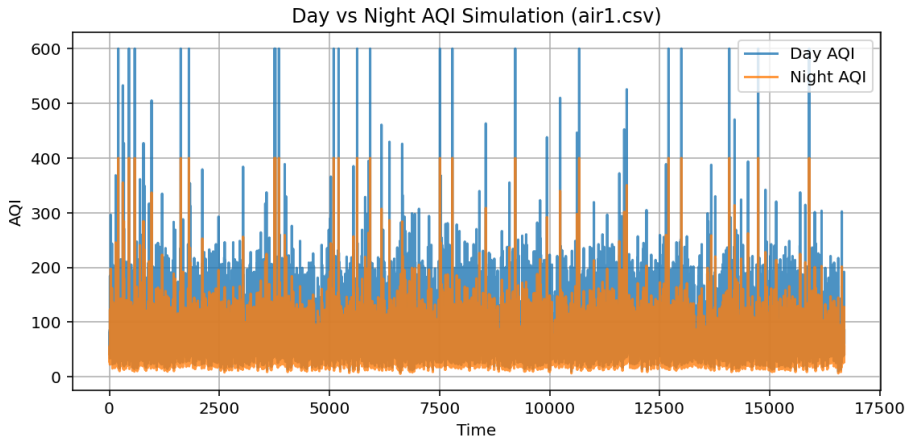


Figure 3: Day vs Night AQI Simulation

This graph compares simulated AQI levels during daytime and nighttime conditions. Daytime values generally appear higher and more variable, while nighttime levels are lower and more stable. This difference reflects the influence of solar radiation and human activity during the day, which enhance photochemical reactions and increase pollutant formation. At night, the absence of sunlight reduces these reactions, leading to relatively lower AQI levels. The simulation captures this diurnal variation effectively, showing that the model accounts for time-dependent atmospheric processes that influence pollution formation and dispersion.

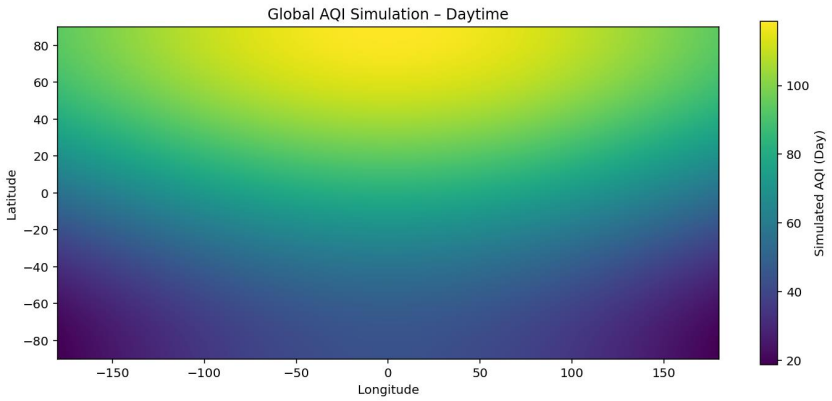


Figure 4: Global AQI Simulation – Daytime

The global daytime heatmap displays simulated AQI variation across latitude and longitude. The distribution shows gradual spatial gradients rather than abrupt changes, with certain latitude bands experiencing higher pollution intensity. This pattern reflects large-scale atmospheric influences such as solar energy distribution, climate zones, and global circulation patterns. The results suggest that the simulation effectively represents broad geographic trends in air quality during daytime conditions, highlighting how environmental factors vary across the planet.

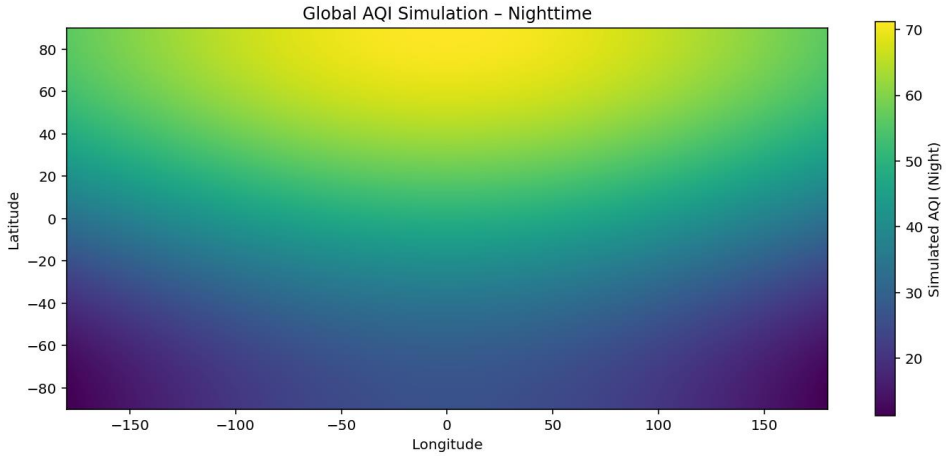


Figure 5: Global AQI Simulation – Nighttime

The nighttime global heatmap shows a similar spatial pattern to the daytime map but with noticeably lower AQI values overall. This reduction reflects the decrease in photochemical activity after sunset, which limits the formation of certain pollutants. Although the geographic distribution remains consistent, the overall pollution intensity is reduced, indicating that the model captures the impact of day–night atmospheric dynamics on global air quality levels.

CONCLUSION

The simulation results demonstrate that the hybrid air quality modelling framework effectively captures the key dynamics of atmospheric pollution across time, space, and environmental conditions. The model successfully represents overall AQI trends, spatial clustering of pollutants, and clear differences between daytime and nighttime air quality. These patterns confirm that the simulation can reproduce realistic environmental behaviour such as pollutant accumulation, dispersion, and photochemical variation. Although the model slightly underestimates extreme pollution spikes and highly localized emission events, it provides a strong representation of general air quality behaviour at regional and global scales. This makes it a reliable analytical tool for understanding long-term pollution patterns and environmental interactions.

The spatial analysis further highlights how pollutants are unevenly distributed, forming concentration hotspots influenced by emission sources and atmospheric transport. The global maps demonstrate how air quality varies across geographic regions, while the diurnal simulation confirms the role of sunlight and human activity in pollution formation. Together, these results show that the model is capable of integrating meteorological, chemical, and spatial factors into a unified framework for air quality prediction and analysis.

Air quality simulation plays an important role in real-world environmental management and public health protection. By modelling pollution levels across time and space, it enables governments and environmental agencies to continuously monitor air quality conditions and anticipate harmful pollution episodes before they occur. This predictive capability supports the development of early warning systems that alert the public during periods of poor air quality, helping reduce health risks. In addition, simulation models identify pollution hotspots—areas where contaminant levels are consistently high—allowing authorities to implement targeted control measures such as traffic restrictions, emission regulations for industries, and improved environmental policies to reduce overall pollution exposure.

These simulations are also highly valuable for urban planning and infrastructure development. They help planners understand how different features of the built environment—such as building density, road networks, and population concentration—affect how pollutants disperse and accumulate. With this knowledge, cities can be designed or modified to improve air circulation and reduce pollution exposure. For health researchers, air quality models provide estimates of how much pollution people are exposed to over time, which is essential for studying the links between air pollution and diseases such as asthma, lung disorders, and cardiovascular conditions. This helps in designing public health interventions and preventive strategies.

Furthermore, air quality simulations support climate and environmental research by explaining how pollution interacts with weather patterns, seasonal changes, and atmospheric chemical processes. Scientists can use these models to understand how temperature, wind, humidity, and sunlight influence pollutant formation and transport. This knowledge is crucial for developing long-term environmental policies, assessing the effectiveness of emission reduction strategies, and planning sustainable development initiatives that balance economic growth with environmental protection.

In addition to practical applications, the modelling framework helps address several major challenges. One of the most significant is the limited availability of physical monitoring stations, which cannot cover every location. Simulations help fill these gaps by estimating pollution levels in areas where direct measurements are not available. They also support health risk assessment by identifying populations that are more vulnerable to pollution exposure, enabling preventive healthcare planning and targeted interventions.

The models are also useful for managing urban environments by identifying major emission sources and analysing how pollutants move through cities. Policymakers can test potential environmental regulations within the simulation before applying them in reality, reducing uncertainty and improving decision-making. In emergency situations such as industrial accidents, wildfires, or extreme weather events, simulations can predict how pollutants will spread, helping authorities respond quickly and effectively. Finally, these models support long-term forecasting by projecting how air quality may change under different climate scenarios, development patterns, or policy interventions, making them essential tools for sustainable environmental planning.